

EXPERIMENT NO 3

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Div:- SY3

Batch:- C3

INPUT CODE

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#include<stdio.h>
void bubble(int a[], int n) {
    int i, j, temp;
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(a[j] > a[j+1]) {
                temp = a[j];
                a[j] = a[j+1];
                a[j+1] = temp;
            }
        }
    }
    printf("\nThe sorted array is : ");
    for(i = 0; i < n; i++) {
        printf("\t%d", a[i]);
    }
}
void insert(int a[], int n) {
    int i, j, temp;
    for(i = 1; i < n; i++) {
        temp = a[i];
        j = i - 1;
        while(j >= 0 && a[j] > temp) {
            a[j+1] = a[j];
            j = j - 1;
        }
        a[j+1] = temp;
    }
}
```

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    }

    printf("\nThe sorted array is : ");
    for(i = 0; i < n; i++) {
        printf("\t%d", a[i]);
    }
}

void selection(int a[], int n) {
    int i, j, min, temp;

    for(i = 0; i < n - 1; i++) {
        min = i;
        for(j = i + 1; j < n; j++) {
            if(a[j] < a[min])
                min = j;
        }
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }
    printf("\nThe sorted array is : ");
    for(i = 0; i < n; i++) {
        printf("\t%d", a[i]);
    }
}

void merge(int a[], int low, int mid, int high) {
    int i, j, k;
    int b[100];
    i = low;
    j = mid + 1;
    k = low;
    while(i <= mid && j <= high) {
        if(a[i] < a[j]) {
            b[k] = a[i];
            i++;
        }
        else {

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        b[k] = a[j];
        j++;
    }
    k++;
}
while(i <= mid) {
    b[k] = a[i];
    i++;
    k++;
}
while(j <= high) {
    b[k] = a[j];
    j++;
    k++;
}
for(i = low; i <= high; i++) {
    a[i] = b[i];
}
}

void mergeSort(int a[], int low, int high) {
    int mid;
    if(low < high) {
        mid = (low + high) / 2;
        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);
        merge(a, low, mid, high);
    }
}

int partition(int a[], int low, int high) {
    int pivot, i, j, temp;
    pivot = a[low];
    i = low + 1;
    j = high;
    while(i <= j) {
        while(i <= high && a[i] <= pivot)
            i++;
        while(a[j] > pivot)

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        j--;

        if(i < j) {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
    }
    temp = a[low];
    a[low] = a[j];
    a[j] = temp;
    return j;
}

void quickSort(int a[], int low, int high) {
    int p;
    if(low < high) {
        p = partition(a, low, high);
        quickSort(a, low, p - 1);
        quickSort(a, p + 1, high);
    }
}

int main() {
    int i;
    int choice;
    int n;
    printf("\nEnter the size of array: ");
    scanf("%d", &n);
    int a[n];
    printf("\nEnter the %d elements : ", n);
    for(i = 0; i < n; i++) {
        printf("\nEnter element %d: ", i + 1);
        scanf("%d", &a[i]);
    }
    printf("\n\n===Methods Available to Sort===");
    printf("\n1. Bubble Sort");
    printf("\n2. Insertion Sort");
    printf("\n3. Selection Sort");

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printf("\n4. Merge Sort");
printf("\n5. Quick Sort");
printf("\n6. Exit");
do {
printf("\n\nEnter your choice: ");
scanf("%d", &choice);
switch(choice) {
case 1:
bubble(a, n);
break;
case 2:
insert(a, n);
break;
case 3:
selection(a, n);
break;
case 4:
mergeSort(a, 0, n - 1);
printf("\nThe sorted array is : ");
for(i = 0; i < n; i++) {
printf("\t%d", a[i]);
}
break;
case 5:
quickSort(a, 0, n - 1);
printf("\nThe sorted array is : ");
for(i = 0; i < n; i++) {
printf("\t%d", a[i]);
}
break;
case 6:
printf("\nExiting Program....\n");
break;
default:
printf("\nEnter a valid choice.");
break;
}
}

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    } while(choice != 6);

    return 0;
}
```

OUTPUT

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Enter the size of array: 6

Enter the 6 elements :
Enter element 1: 43

Enter element 2: 34

Enter element 3: 56

Enter element 4: 76

Enter element 5: 2

Enter element 6: 31

===Methods Available to Sort===
1. Bubble Sort
2. Insertion Sort
3. Selection Sort
4. Merge Sort
5. Quick Sort
6. Exit

Enter your choice: 1

The sorted array is :   2       31       34       43       56       76

Enter your choice: 2

The sorted array is :   2       31       34       43       56       76

Enter your choice: 3

The sorted array is :   2       31       34       43       56       76

Enter your choice: 4

The sorted array is :   2       31       34       43       56       76
```

Enter your choice: 5

The sorted array is : 2 31 34 43 56 76

Enter your choice: 6

Exiting Program....

...Program finished with exit code 0

Press ENTER to exit console.